

MAKTUB II

Trefoiled state,
symbolic learning
and just-in-time growth

The five-star log-polar field, doubly packed pinions,
the W timeline and deterministic structural rewriting.

MAKTUB $\rightarrow$ TOMK $\rightarrow$ TOM-GROK

5×5 LOCAL FIELD $\rightarrow$ L-SYSTEM / JUST IN TIME

$$\begin{aligned}x_{n+1} &= \text{Act}(x_n, d_n, e_n), \\d_{n+1} &= \text{Rewrite}(x_n, d_n, e_n).\end{aligned}$$

The event writes its successor state and next governing description. The local structure required by the event is materialized through that same written execution.

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A new formal specification following the supplied development sequence.

State, learning, execution, topology, fields and growth in one record.

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Reading convention. $\coloneqq$ introduces a definition; $=$ states equality in that definition; $\rightsquigarrow$ assigns a domain reading. Source locators refer to the supplied PDF page numbers. The concrete address map, grammar and transaction protocol are the definitions of this edition.

The previous specification supplies the paired kernel, mirror, external anchor and two-field representation. The new development supplies the trefoil, five-by-five site structure, W timeline, paired Klein/SDF packing and just-in-time growth. They are composed below in that order.

01 / FORMAL PROGRESSION

The progression and its new closure

Core declaration. Maktub is extended from paired state-and-rule bookkeeping into a trefoiled execution architecture whose local field can grow under its own committed description. The just-in-time addition makes the required structure part of the transition itself.

The development sequence

Layer	Written object	Contribution to the progression
Maktub	(x_n, d_n) and the paired update	State and future governance are recorded together.
TOMK	Explicit hypotheses and evaluation records	Candidate descriptions acquire accepted, rejected and ambiguous outcomes.
TOM-GROK	Typed operator field and runtime interpretation	The state and symbolic descriptions meet an executable instruction stream.
Five-star field	25 addressed sites with packed pinions	Location, scale, field values and local transformations share a finite module.
L-system / JIT	Guarded productions and demand events	The next required module is realized before its first field access.

What the addition changes

The earlier kernel asks: *What state and rule follow this event?* The new closure also asks: *Which local structure must exist for this event to be executed?* The answer is a deterministic materialization record generated from the current state, rule and input.

Let $\mathcal{Y}_n$ denote the event's work record: requested addresses, grammar derivations, field reads, numerical stages and symbolic evidence. Then the enclosing progression is

$$(x_n, d_n, e_n) \mapsto \mathcal{Y}_n \mapsto (x_{n+1}, d_{n+1}). \quad (1.1)$$

The work record is derived under d_n . The resulting d_{n+1} governs the following event. This preserves the original paired update while extending its state to include the grown structure.

The trefoiled reading

State, symbolic rule evaluation, execution are the three mutually connected roles. The five-star field and the growth grammar are carried through these roles: the field is stored as state, its productions are written as rules, and materialization is performed by execution.

02 / FORMAL PROGRESSION

The inherited paired kernel

Let X be the state space, D the space of operator descriptions and E the admitted event space. A committed configuration is $s_n = (x_n, d_n) \in X \times D$.

$$\mathcal{K}(s_n, e_n) := (\text{Act}(x_n, d_n, e_n), \text{Rewrite}(x_n, d_n, e_n)). \quad (2.1)$$

Same-snapshot rule. Both outputs are functions of the same pre-transition triple (x_n, d_n, e_n) . Every intermediate result is derived from that triple and retained in the event's work record.

Six commitments carried forward

Written execution. The instruction belongs to the recorded configuration.

Paired change. The state transition and operator rewrite are committed as one event.

Located operation. Every local read has a declared origin, chart and address.

Explicit passage. Retention, release and transfer are named branches of the operator.

Mirror. The elementary involution is recorded as an operation, with its own type.

Continuation. The output supplies the next state and the description of its next passage.

The one-bit reduction

With $B = \{0, 1\}$, define

$$\mathcal{M}(b) = 1 - b, \quad \mathcal{M}(\mathcal{M}(b)) = b. \quad (2.2)$$

In the reduced presentation, $x_n \equiv d_n \equiv b_n$ and the paired mirror step is $(x_{n+1}, d_{n+1}) \equiv (\mathcal{M}(b_n), \mathcal{M}(b_n))$. In the extended presentation, b_n is an orientation or selection component of a larger typed state.

The external anchor

The inherited capstone is

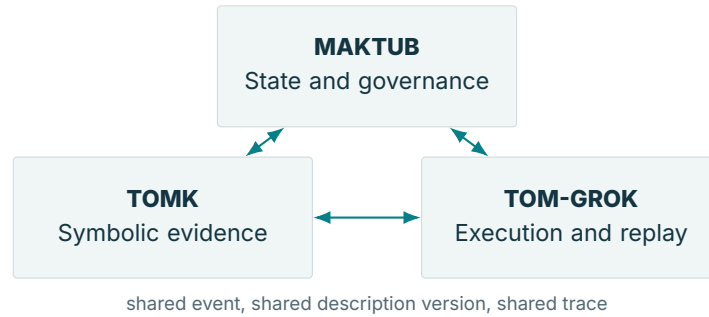
$$M = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad \mathcal{C} = (M, a_{\text{out}}, \text{lower-left}). \quad (2.3)$$

The external entry a_{out} remains the designated output destination. In the growth extension, an outlet can reference a successor module through an explicit transfer record. The anchor is kept in control metadata; the local 25-site table counts its payload sites separately.

03 / FORMAL PROGRESSION

The trefoiled execution structure

Three roles, one committed event



The extended state

Use the following explicit record:

$$x_n := (z_n, \omega_n, \mathcal{G}_n, \mathcal{P}_n, \mathcal{L}_n, \mathcal{R}_n, \mathcal{T}_n). \quad (3.1)$$

$z_n = (p_n, W_n, \xi_n)$	Position, timeline coordinate and additional dynamic channels.
ω_n	Current parametric grammar word, including dormant growth markers.
$\mathcal{G}_n$	Materialized module directory and declared bond graph.
$\mathcal{P}_n$	Packed pinion records attached to materialized sites.
$\mathcal{L}_n$	Field samples, address indices and validity metadata.
$\mathcal{R}_n$	Hypothesis outcomes, counterexamples and comparison records.
$\mathcal{T}_n$	Ordered event, transfer and commit trace.

The operator description

The description d_n contains the instruction tape, grammar productions, coordinate profiles, field definitions, gate functions, hypothesis family, selection rule and arithmetic profile. The interpreter reads these descriptions as typed inputs.

The word *emulsion* denotes this composition of the state kernel and symbolic learner within a shared execution medium. *Trefoiled* names their three-role incidence structure. The same event traverses all three roles and returns a single paired successor.

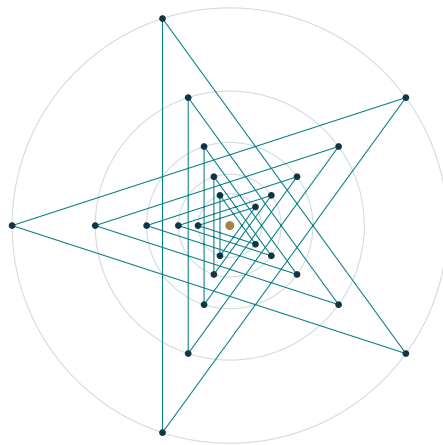
04 / FORMAL PROGRESSION

Five pentagrams and twenty-five sites

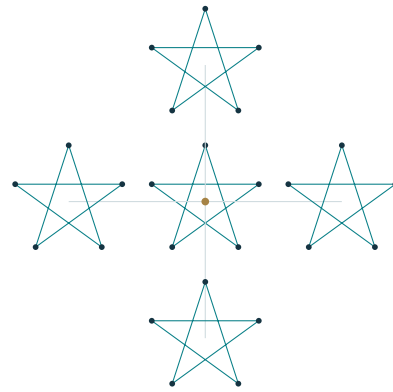
The elementary payload module is indexed by

$$V_{25} := \{v_{ij} : i, j \in \{0, 1, 2, 3, 4\}\}, \quad |V_{25}| = 25. \quad (4.1)$$

A fixed i identifies one five-site group. Within each group, the pentagram incidence is $j \leftrightarrow j + 2 \pmod{5}$. Connecting adjacent radial groups supplies the inter-group routing relation.



concentric log-polar realization



hub-and-four display profile

One identity set, two presentations

The concentric realization gives each group a geometric radius and each site an angular slot. The hub-and-four profile retains the source's central pentagram with four surrounding pentagrams. A display map changes where a site is drawn; its identity v_{ij} and stored payload remain the same.

Counting rule. The 25 objects are the labeled payload sites. The origin, module headers, dormant buds and external anchor are control records. Crossings of drawn pentagram edges are geometric intersections, with site membership determined by the explicit identity set.

A larger field consists of addressed copies V_{25}^v , where v is a module path. The module remains the elementary 5×5 unit as the grammar extends the collection.

05 / FORMAL PROGRESSION

Log-polar addressing and binary search

The coordinate map

For a planar chart centered at o , let $q = (q_1, q_2) = p - o$ and set

$$r = \sqrt{q_1^2 + q_2^2}, \quad \rho = \ln(r/r_{\min}), \quad \theta = \text{atan2}(q_2, q_1) \pmod{2\pi}. \quad (5.1)$$

The finite module uses $r_{\min} > 0$, $\Delta\rho > 0$ and $r_{\max} = r_{\min}e^{5\Delta\rho}$. Its radial domain is $r \in [r_{\min}, r_{\max})$; the core and exterior have named routing branches.

$$i = \left\lfloor \frac{\rho}{\Delta\rho} \right\rfloor, \quad j = \left\lfloor \frac{5\theta}{2\pi} \right\rfloor, \quad k = 5i + j \in \{0, \dots, 24\}. \quad (5.2)$$

The stored representative of a cell is its log-radial and angular midpoint:

$$r_i = r_{\min}e^{(i+1/2)\Delta\rho}, \quad \theta_j = \frac{2\pi(j+1/2)}{5}, \quad p_{ij} = o + r_i(\cos \theta_j, \sin \theta_j). \quad (5.3)$$

The binary decision structure

Build a balanced binary search tree over the ordered keys $0, \dots, 24$. For an inclusive interval $[a, b]$, its root key is $m = \lfloor (a + b)/2 \rfloor$, with child intervals $[a, m - 1]$ and $[m + 1, b]$. An empty interval has no node.

A query compares its key with m : equality selects the payload, a smaller key selects the left child, and a larger key the right child. For the complete 25-key tree the longest root-to-site path contains five visited nodes.

Functional composition. The fivefold grouping defines the local field; the log-polar map defines its cell key; the two-way comparator defines the search path. Their composition is the log-encoded polar LUT binary search architecture.

Origin and seam

The origin is the chart's exact reference point and follows the core route. Angular reduction fixes the $0/2\pi$ seam, while half-open cell intervals assign every interior boundary to one cell. A global address is (v, χ, k) : module path, chart identifier and local key. The resolver uses the profile's declared chart-ownership order, with lexicographic address order to settle equal-priority candidates.

06 / FORMAL PROGRESSION

Spatial charts and eigenvector axes

The committed frame

Let the geometry profile contain a real symmetric matrix A_n and an ordered orthonormal eigenframe E_n :

$$A_n E_n = E_n \Lambda_n, \quad E_n^\top E_n = I, \quad q = E_n^\top (p - o_n). \quad (6.1)$$

The matrix, eigenvalue order, signs and basis choices within repeated eigenspaces are stored with the frame. This gives the event a reproducible set of axes. The current event uses E_n throughout; a rewritten frame belongs to the next description.

The spherical-sector reading

For a three-dimensional spatial chart, use

$$p = o_n + r E_n \begin{pmatrix} \cos \beta \cos \theta \\ \cos \beta \sin \theta \\ \sin \beta \end{pmatrix}, \quad r > 0, \quad -\frac{\pi}{2} \leq \beta \leq \frac{\pi}{2}. \quad (6.2)$$

A *sphere fracture* is a named sector with a radial interval, an azimuth interval and an elevation interval. The 25 local keys index the five radial and five azimuth subdivisions of a chosen chart; the chart identifier carries the elevation band or alternate coordinate patch.

At a pole, the chart profile supplies its canonical azimuth. Additional patches are addressed through the same directory. The common origin is fixed by o_n , rather than inferred from a displayed scalar value.

The raster is a readout

A renderer is an explicit map

$$\Pi_{\text{view}} : (p, W, \xi, \nu, \chi) \mapsto \text{screen position and displayed channels}. \quad (6.3)$$

It selects a camera, slice or projection of the underlying state. The full record retains spatial coordinates, time, chart identity, orientation and field values after the image is generated.

Axis commitment. Geometry is evaluated in the recorded frame, transport is transformed with that frame, and rendering reads the resulting state. A frame update is an explicit part of operator rewriting.

Source base [S2], pp. 5–8; [S0], p. 6. The eigenframe convention is defined here; the real symmetric eigenbasis construction is supported by [M3].

07 / FORMAL PROGRESSION

The W timeline and RK4

The dynamic state is $z = (p, W, \xi) \in \mathbb{R}^{4+q}$, with $p \in \mathbb{R}^3$, a scalar timeline coordinate W , and q additional channels. Write the vector field as

$$\frac{dz}{d\tau} = F_{d_n}(z, e_n), \quad \frac{dW}{d\tau} = 1. \quad (7.1)$$

The four evaluation stages

For an event step $h_n > 0$, keep the governing description fixed and evaluate

$$\begin{aligned} k_1 &= F_{d_n}(z_n, e_n), \\ k_2 &= F_{d_n}(z_n + \frac{h_n}{2}k_1, e_n), \\ k_3 &= F_{d_n}(z_n + \frac{h_n}{2}k_2, e_n), \\ k_4 &= F_{d_n}(z_n + h_n k_3, e_n), \\ z^* &= z_n + \frac{h_n}{6}(k_1 + 2k_2 + 2k_3 + k_4). \end{aligned} \quad (7.2)$$

The temporal components of all four stages equal one, giving

$$W^* = W_n + h_n. \quad (7.3)$$

Thus W is the fourth *coordinate* of the spatial-temporal presentation; k_4 is the fourth *stage vector* used to produce the successor. Both have explicit, separate places in the record. The interpreter writes W^* by direct timeline addition in the declared number format.

Adding operators and channels

An extended operator tape can contain any declared finite composition

$$\mathcal{O}_{d_n} = O_m \circ \dots \circ O_2 \circ O_1, \quad O_j : U_j \rightarrow U_{j+1}. \quad (7.4)$$

Its length m , the state-channel count q and the integration-stage count are separate parameters. New channels extend z and F ; new typed operators extend the tape. RK4 remains the four-stage integrator for the chosen state vector.

Demand during a temporal step

Every evaluation of F_{d_n} performs its required field reads through the same just-in-time access procedure. The stage inputs, addresses read, chart transports and outputs are retained in $\mathcal{Y}_n$ before the paired commit.

Source base [S2], pp. 6–9; [S3], pp. 5–6. RK4 stage algebra: [M2]. The augmented-time state and frozen-description scheduling are defined here.

08 / FORMAL PROGRESSION

Two Klein channels in every pinion

The doubled topological carrier

Each pinion carries two labeled channels, $\mathbb{K}^A$ and $\mathbb{K}^B$. Define either carrier by the square quotient

$$\mathbb{K} := [0, 1]^2 / \sim, \quad (0, v) \sim (1, v), \quad (u, 0) \sim (1 - u, 1). \quad (8.1)$$

Equivalently, the covering-plane generators are

$$a(u, v) = (u + 1, v), \quad b(u, v) = (-u, v + 1). \quad (8.2)$$

The periodic identification and reflected identification specify how a trajectory continues between chart representatives.

Canonical coordinates and orientation

For a covering coordinate (u, v) , define

$$m = \lfloor v \rfloor, \quad \text{Norm}(u, v) = (\text{fract}((-1)^m u), v - m). \quad (8.3)$$

The associated tangent transformation is

$$J_m = \begin{pmatrix} (-1)^m & 0 \\ 0 & 1 \end{pmatrix}, \quad \varepsilon^* = \varepsilon \oplus (m \bmod 2). \quad (8.4)$$

Here $\varepsilon \in \{0, 1\}$ records local orientation and $\oplus$ is exclusive OR. Vectors are transported by J_m when their chart representative changes. Scalar fields agree on identified points; vector fields transform by the corresponding chart derivative.

The paired pinion record

The topological part of a site record is

$$(\mathbb{K}^A, \mathbb{K}^B, \kappa^A, \kappa^B, \varepsilon^A, \varepsilon^B, J^A, J^B). \quad (8.5)$$

A declared pair-swap operation $S(a, b) = (b, a)$ satisfies $S^2 = I$. It supplies a doubled-channel mirror compatible with the inherited one-bit presentation.

Continuity rule. A chart exit is followed by its recorded identification and tangent transport. The trace retains the representative change, orientation update and destination channel.

09 / FORMAL PROGRESSION

The paired signed-distance fields

Two explicitly bounded regions

For each pinion, declare two regular closed field regions $\Omega_A, \Omega_B \subset \mathbb{R}^m$ with nonempty boundaries. They supply the metric gate profiles associated with its two channels. For $\sigma \in \{A, B\}$, define

$$\delta_\sigma(y) = \begin{cases} -\text{dist}(y, \partial\Omega_\sigma), & y \in \text{int}(\Omega_\sigma), \\ 0, & y \in \partial\Omega_\sigma, \\ +\text{dist}(y, \partial\Omega_\sigma), & y \notin \Omega_\sigma. \end{cases} \quad (9.1)$$

Each field record declares its region, coordinate map, units and sign convention. The Klein channel supplies the topological transport; the region Ω_σ supplies the inside/boundary/outside classification.

Sampling in a channel

A profile map χ_σ maps channel state to the field's evaluation coordinates. The paired read is

$$f(v, z) = \begin{pmatrix} \delta_A(\chi_A(v, z)) \\ \delta_B(\chi_B(v, z)) \end{pmatrix}. \quad (9.2)$$

When χ_σ is defined on $\mathbb{K}^\sigma$, its values respect the quotient identifications. Chart-local definitions carry their matching transition maps in the pinion record.

A concrete gate profile

For a positive gate width η_σ , this edition's continuous selection gate is

$$g_\sigma(y) = \text{clip}\left(\frac{1}{2} - \frac{\delta_\sigma(y)}{2\eta_\sigma}, 0, 1\right), \quad \text{clip}(a, 0, 1) = \min(1, \max(0, a)). \quad (9.3)$$

The value selects full retention, a transition interval or release according to the declared field sign and width. A Boolean gate uses an explicitly stated threshold instead.

Field representation modes

The packed type distinguishes an analytic signed-distance field, sampled signed-distance values, and an implicit contour field. In contour mode, a zero set defines the region. A distance-mode realization then applies the distance definition above to that region. The storage record names the mode used by every gate.

Source base [S0], pp. 16–17; [S2], pp. 9–13. The two-region pairing, pullback read and gate profile are defined here.

10 / FORMAL PROGRESSION

Double Hadamard and the field mixture

Two successive elementwise products

Let the local payload be $U \in \mathbb{C}^{25 \times c}$ and let the two gate arrays $G^{(1)}, G^{(2)}$ have the same shape. Define

$$Y = U \odot G^{(1)}, \quad Z = Y \odot G^{(2)}, \quad Z_{ab} = U_{ab} G_{ab}^{(1)} G_{ab}^{(2)}. \quad (10.1)$$

These are the two elementwise Hadamard operations in the doubly packed pinion. The first gate can encode mixture or distribution weights; the second can encode paired boundary gates, orientation signs or phase factors. Their meanings are recorded in the operator description.

The two-field specialization

For the inherited two-component field, choose

$$U = (F_A, F_B), \quad G^{(1)} = (1 - \alpha, \alpha), \quad G^{(2)} = (g_A, g_B). \quad (10.2)$$

The summed output is

$$F_{\text{out}} = (1 - \alpha)g_A F_A + \alpha g_B F_B. \quad (10.3)$$

Taking $g_A = g_B = 1$ recovers the previous weighted mixture. Its quantity-to-weight bridge remains

$$\alpha(N) = \text{clip}\left(\frac{9 - \log_{10} N}{4}, 0, 1\right), \quad N > 0. \quad (10.4)$$

The capital N distinguishes this model quantity from the event index n .

The phase transform retains its own symbol

The inherited normalized Hadamard matrix is

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad H^2 = I. \quad (10.5)$$

When the tape includes phase mixing, H acts by matrix multiplication on a two-component channel vector. The symbol $\odot$ continues to denote elementwise multiplication. Operator order, channel shape and the choice of phase or gate operation are explicit tape fields.

Pinion action. Read the paired fields, transport their channel state, apply the two declared elementwise gates, and execute any separately declared phase or output-mixing operation.

11 / FORMAL PROGRESSION

The lowercase phi and psi distribution

The constants in the new topology layer

Use the latest source's positive reciprocal pair:

$$\varphi_g := \frac{1 + \sqrt{5}}{2}, \quad \psi_g := \frac{1}{\varphi_g} = \varphi_g - 1. \quad (11.1)$$

Thus $\varphi_g \psi_g = 1$, $\varphi_g^2 = \varphi_g + 1$, and the module's radial step is $\Delta\rho = \ln \varphi_g$. The successive boundaries are

$$r_i^{\text{edge}} = r_{\min} \varphi_g^i, \quad i = 0, \dots, 5. \quad (11.2)$$

A step in log radius is therefore the fixed increment $\ln \varphi_g$, while the corresponding spatial radius changes by a factor φ_g .

Outward scale and local refinement

The distribution profile has two named operations:

$$\text{Outward}(r) = \varphi_g r, \quad \text{Refine}(s) = \psi_g s. \quad (11.3)$$

A parent site (v, i, j) can carry a child module with center $a_{v'} = p_{v,ij}$ and scale $s_{v'} = \psi_g s_v$, where $v' = v\|(i, j)$. The child's frame is the parent frame composed with its declared branch rotation. The module path preserves parentage even when projected drawings overlap.

Time placement is written separately

A preview lead time can use the same scale family:

$$\tau_{\text{lead}}(d) = \tau_0 \psi_g^d, \quad \tau_0 > 0, \quad (11.4)$$

where d is module depth. This is a scheduling parameter of the profile; W remains measured in the declared time unit. Spatial scale, angular phase and temporal lead each retain their own typed field.

Continuity with the earlier notation

The previous specification uses the labels $\Phi \approx 1.618$ and lowercase $\phi \approx 0.618$. The correspondence register for this edition is

$$\Phi_{\text{previous}} \rightsquigarrow \varphi_g, \quad \phi_{\text{previous}} \rightsquigarrow \psi_g. \quad (11.5)$$

The numerical values are fixed by the explicit definitions at the top of this page. The new lowercase pair is recorded as φ_g, ψ_g throughout the growth layer.

12 / FORMAL PROGRESSION

The deterministic growth grammar

Alphabet, parameters and seed

Define a parametric grammar $\mathfrak{G} = (\Sigma, \omega_0, \mathcal{P}_L, I)$. Its alphabet is $\Sigma = \{T, H, B, P, A\}$: module seed, header, five-site group, payload site and dormant child bud. The initial word is $\omega_0 = T(\epsilon)$, where ϵ is the root path. I interprets a completed module as its site, pinion and lookup records.

The five-by-five productions

For a demanded module v , the productions are

$$\begin{aligned} T(v) &\longrightarrow H(v) \prod_{i=0}^4 B(v, i), \\ B(v, i) &\longrightarrow \prod_{j=0}^4 [P(v, i, j) A(v, i, j)]. \end{aligned} \tag{12.1}$$

Here $\prod$ denotes ordered string concatenation in increasing index order. The first derivation exposes five groups. The second replaces those groups in parallel with 25 payload sites and their dormant successor markers.

The recursive continuation

When the demand set includes the child path $v\|(i, j)$,

$$A(v, i, j) \longrightarrow T(v\|(i, j)). \tag{12.2}$$

The child inherits its placement parameters from the distribution profile. Headers and realized payloads use identity productions; an undemanded seed or bud also retains itself. Each guard selects exactly one production, including the identity branch.

Parallel rewrite rule. One derivation pass evaluates all guards against the same pre-pass word and demand set, then replaces all selected modules simultaneously. Newly written symbols participate in the next pass.

Realization

The interpreter realizes a requested module after its header and all 25 payload symbols are present. It assigns unique addresses, fills both field channels, installs the paired Klein transport, installs both gate arrays, and builds the local key index. The dormant buds remain control symbols. Realization installs each module identity once; later passes preserve its existing payload.

The exact productions above are the concrete L-system supplied by this edition to express the source's growth requirement. A production-set change is written into d_{n+1} and first governs the next event.

Source base [S2], pp. 16–18; [S3], pp. 5–6. Parametric and parallel grammar conventions: [M1], chapter 1, sections 1.2 and 1.10. The productions are defined here.

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Just-in-time materialization

An address exists before its payload is allocated

The seed and coordinate profile determine a module's procedural identity, position and child addresses. A payload is realized when an event first requires that identity. This separates the description of possible structure from the materialized working set.

Let $\mathcal{D}_n$ be the addresses requested by the event and let $\overline{\mathcal{D}}_n$ include their ancestor modules and declared sampling dependencies. For a read a at timeline position W , define

$$\text{Need}(a, W) = 1 \implies \text{Ready}(a, W) = 1 \text{ before the read is issued.} \quad (13.1)$$

The demand-read procedure

A field access first resolves its module and local key. It then adds the address, ancestor modules and interpolation stencil to the demand set. The interpreter applies the guarded grammar until those records are ready, then performs the read under the same description version.

The realized set evolves as

$$\mathcal{V}^{\text{work}} = \mathcal{V}_n \cup \text{Realize}_{d_n}(\overline{\mathcal{D}}_n). \quad (13.2)$$

Identity is assigned by module path and local key. Repeated requests reuse the same record. Parent modules are realized before descendants, and equal-priority requests are ordered lexicographically by address.

Preview and immediate demand

A declared trajectory preview can request the modules needed over

$$W \in [W_n, W_n + h_n + \tau_{\text{lead}}]. \quad (13.3)$$

RK4 stage reads also issue immediate demands as their stage coordinates become available. Preview requests and stage requests share the same directory, grammar and readiness rule.

The just-in-time addition. Growth is scheduled by the event's required support. The written grammar supplies what appears; the address profile supplies where it appears; the timeline supplies when it becomes available for execution.

The work record stores every request, production pass, new module and first-use time. Materialized structure is incorporated into x_{n+1} alongside the resulting numerical and symbolic state.

Source base [S3], pp. 5–6; [S2], pp. 16–18. The demand closure, readiness rule and event-local working set are the scheduling definition of this edition.

14 / FORMAL PROGRESSION

Bonds, diffusion and dissipation

A written bond field

Let the active-site graph have symmetric bond weights $b_{ij} = b_{ji} \geq 0$. Its Laplacian is

$$(L_B u)_i = \sum_j b_{ij}(u_i - u_j). \quad (14.1)$$

The weights encode the declared strength of coupling between sites. A concrete scalar transport profile is

$$\dot{u} = -DL_B u - \Gamma u + s, \quad D \geq 0, \quad \Gamma = \text{diag}(\gamma_i), \quad \gamma_i \geq 0. \quad (14.2)$$

The first term distributes the field through its bonds. The second releases field content into a named sink. The third supplies an admitted source. These are the transport, dissipation and input components of the written profile.

The local and external ledgers

For the site sum $M = \mathbf{1}^\top u$,

$$\dot{M} = -\mathbf{1}^\top \Gamma u + \mathbf{1}^\top s. \quad (14.3)$$

Define an external ledger r by $\dot{r} = \mathbf{1}^\top \Gamma u$ and an input ledger a by $\dot{a} = \mathbf{1}^\top s$. The combined bookkeeping identity is

$$\frac{d}{d\tau}(M + r - a) = 0. \quad (14.4)$$

The state therefore records both what remains in the active field and what crosses its release interface.

Growth and redistribution

When a grammar event refines an active field, use a declared nonnegative transfer matrix Q with $\mathbf{1}^\top Q = \mathbf{1}^\top$:

$$u^{\text{new}} = Qu^{\text{old}}. \quad (14.5)$$

This assigns the parent's field content to its active successors. Structural metadata and historical parent records remain in the directory; the active ledger identifies where the transferred payload now resides.

The two actions. Diffusion names redistribution through the bond graph. Dissipation names release into the external ledger. The L-system supplies the organized receiving structure, and the transfer operator supplies the allocation rule.

The terms *bond*, *growth*, *cell* and *tissue* remain available as domain readings of these explicitly defined relations and modules.

Source base [S0], pp. 7, 13; [S2], pp. 14–18; [S3], pp. 5–6. The graph transport equation and transfer ledgers are this edition's concrete bookkeeping profile.

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Symbolic evidence and operator rewriting

The finite hypothesis record

Let $\mathcal{H}_n = \{h_1, \dots, h_m\}$ be the ordered hypothesis family declared in d_n , and let $\mathcal{B}_n$ be the event's evaluation dataset. Each hypothesis has a typed evaluator, identifier and comparison predicate.

$$\mathcal{R}_n^* = \text{Eval}_{d_n}(\mathcal{H}_n, \mathcal{B}_n) = \{(h_j, \text{status}_j, \text{evidence}_j)\}_{j=1}^m. \quad (15.1)$$

The report records accepted hypotheses, rejected hypotheses and their counterexamples, competing accepted alternatives, and comparisons with the prior recorded evaluation. The source's TOMK snapshot names 121 literal hypotheses, four families and sixteen datasets; those are the supplied snapshot's profile values.

Selection is an explicit function

For the accepted set $\mathcal{A}_n$, define this edition's default selector by

$$\text{Select}(\mathcal{A}_n) = \begin{cases} \min_{\text{id}} \mathcal{A}_n, & \mathcal{A}_n \neq \emptyset, \\ \text{hold}, & \mathcal{A}_n = \emptyset. \end{cases} \quad (15.2)$$

The full accepted set remains in the report when multiple hypotheses satisfy the profile. A different ranking is represented by a different declared selector.

The rewrite belongs to the next event

Let $\mathcal{Y}_n$ include the materialization trace, field outputs and completed evidence report. The successor description is

$$d^* = \text{RewriteFrom}_{d_n}(x_n, e_n, \mathcal{Y}_n, \text{Select}(\mathcal{A}_n)). \quad (15.3)$$

The instruction tape specifies which fields of d_n are retained, replaced or extended. A selected hypothesis can update a gate profile, production parameter or subsequent operator choice through its typed encoding.

Evidence-to-rule link. The current rule determines how evidence is read and how the next description is written. The newly selected description is committed with the successor state and first applies to the following event.

The trefoil closes

Maktub records the paired successor. TOMK supplies the symbolic outcome record. The TOM-GROK execution role interprets the instructions that produce both. The local field and growth history remain available for replay within that same event record.

Source base [S0], p. 4; [S1], pp. 2–4; [S3], pp. 2–5. Snapshot counts are recorded from [S0]; the selector and typed rewrite interface are defined here.

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The complete event transaction

The event is evaluated in a working copy derived from the committed snapshot. Grammar passes, field reads and numerical stages all use d_n ; the next description is written after their results have been assembled.

```

TRANSITION(x, d, e):
    work := snapshot(x)
    trace := begin_event(version(d), e)
    demands := declared_requests(x, d, e)

    DEMAND_READ(address, W):
        add_dependency_closure(demands, address, d)
        while any requested module is not ready:
            word := parallel_rewrite(work.word, demands, d)
            work.word := word
            realize_newly_completed_modules(work, word, d)
            record_derivation(trace, word)
        return read_typed_payload(work, address, d, W)

    result := interpret_tape(work, d, e, DEMAND_READ)
    # RK4 stages and pinion operations use DEMAND_READ.
    report := evaluate_hypotheses(x, d, e, result, trace)
    next_d := rewrite_description(x, d, e, result, report)
    next_x := assemble_state(work, result, report, trace)

    if contract(next_x, next_d, trace, d):
        return atomic_commit(next_x, next_d)
    else:
        return record_rejected_event(x, d, e, trace)

```

The commit contract

The contract requires unique site identities; ready dependencies before reads; one description version throughout current-event execution; matched channel shapes and field units; recorded chart transports; and a complete paired output. The rule also checks that $W^* = W_n + h_n$ for the declared accepted step.

The explicit rejection branch

A rejected event retains the preceding active field and governing description and appends its diagnostic record. Any repair, retry or alternative route is a subsequent named event. This gives the source's instruction-adherence requirement a precise transaction boundary.

$$\mathcal{K}_{\text{II}}(s_n, e_n) = \begin{cases} (x^*, d^*), & \text{Contract} = 1, \\ (\text{AppendReject}(x_n, e_n, \mathcal{Y}_n), d_n), & \text{Contract} = 0. \end{cases} \quad (16.1)$$

Source base [S0], pp. 4, 20; [S2], pp. 16–18; [S3], pp. 5–6. The pseudocode and atomic commit contract are this edition's integrated execution definition.

Packed data and runtime interpretation

The local module contract

Every realized module has a stable path v , chart identifier, parent pointer, center, frame, scale, radial range, angular convention, grammar version and 25 local site keys. Every site points to a complete pinion record.

Record	Required contents
Geometry	Origin, frame, radial bounds, five-by-five quantizer, chart and branch placement.
Topological pair	Two Klein channel coordinates, orientation bits and chart transport rules.
Field pair	Two region definitions or field references, units, representation mode and sampler.
Operator pair	Two same-shaped Hadamard gate arrays, channel meanings and operation order.
Index	Site key, module path, lookup payload and binary-search child references.
Growth	Seed lineage, production version, dormant child identities and readiness state.
Evidence	Input identity, hypothesis report, selected rewrite and event trace.

The inherited field texture

A four-channel field texture retains the previous layout: R stores F_A , G stores F_B , B is reserved, and A stores validity. The geometric transform and field units are shared by R and G. Klein coordinates, gates and genealogy occupy separate typed records or declared buffers.

The interpreter contract

The source names CPU, CUDA and WebGPU interpretation roles. This edition defines their common semantic target as the same instruction tape, address rule, channel layout, event order and commit record. A runtime profile declares number format, rounding, elementary functions, reduction order, gate evaluation and serialization.

Replay compares canonical decoded records under the named arithmetic profile. The event includes the interpreter and profile version used for the run; the seed and production identities accompany the recorded outputs.

Identity placement. Author attribution belongs to document metadata. The computational payload uses module paths, field identities and event keys. Personal attribution fields are separate from field channels and runtime state.

Source base [S0], pp. 4, 17; [S1], p. 4; [S3], p. 4; [S2], pp. 8–10. The packed schema and interpreter contract are defined here.

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A worked twenty-five-site execution

Declared example profile

Take $r_{\min} = 1$, $\Delta\rho = \ln \varphi_g$, $r_{\max} = \varphi_g^5$, root word $T(\epsilon)$ and a query at

$$r = \varphi_g^{9/4}, \quad \theta = \frac{7\pi}{10}. \quad (18.1)$$

The address equations give

$$i = \lfloor 9/4 \rfloor = 2, \quad j = \lfloor 7/4 \rfloor = 1, \quad k = 5(2) + 1 = 11. \quad (18.2)$$

The event demands the root module. Two grammar passes expose its header, 25 payload sites and dormant child markers. The interpreter installs their paired fields and pinion operators before key 11 is read.

The lookup path

The balanced-tree recurrence gives the comparison path

$$12 \longrightarrow 5 \longrightarrow 8 \longrightarrow 10 \longrightarrow 11. \quad (18.3)$$

Its branches are left, right, right, right, then equality. The result is site $v_{2,1}$ in the root module.

The two products

For this example's stored two-field values and constant gates, set

$$U = (0.2, -0.4), \quad G^{(1)} = (0.5, 0.5), \quad G^{(2)} = (1, 0.75). \quad (18.4)$$

Then

$$Y = (0.1, -0.2), \quad Z = (0.1, -0.15), \quad F_{\text{out}} = -0.05. \quad (18.5)$$

These numbers are in the profile's declared field unit. The mixture weight is $\alpha = 0.5$, which also corresponds to $N = 10^7$ under the inherited bridge.

The timeline step and commit

For an additional scalar channel with $\xi = 0.25$, choose $\xi_n = 1$, $W_n = 5$ and $h_n = 0.1$. All four RK4 stages have ξ component 0.25 and W component 1, giving

$$\xi^* = 1.025, \quad W^* = 5.1. \quad (18.6)$$

The selected symbolic outcome is recorded according to the declared hypothesis profile. The commit contains the realized module, lookup path, gate outputs, temporal stages, evidence report and next description. The next event starts from that paired record.

Source base Worked values are defined and calculated for this edition using sections 5, 7, 10, 12–16. Source structures: [S0], pp. 15–18; [S2], pp. 1–10; [S3], pp. 5–6.

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Continuation and domain correspondences

The shared passage

The inherited sequence remains

$$\text{origin} \rightsquigarrow \text{written operation} \rightsquigarrow \text{boundary passage} \rightsquigarrow \text{successor}. \quad (19.1)$$

The growth addition gives the successor an explicitly realized local support. A parent can release a payload, an external anchor can designate its destination, and a child module can receive the payload through a recorded transfer.

Source vocabulary	Place in the extended formal record
Micro-hard / Macrosoft	Local constraint profile and distributed field presentation.
XNA / DNA; Mono	Named linguistic correspondences in the source's interpretation layer.
Bond, cell, tissue	Coupling relation, local module and connected module collection.
Mirror / Y-pole toggle	Declared involution or orientation-selection bit.
Pinion / terminal outlet	Paired local operator with an explicit output destination.
Tether / secondary pouch	Parent attachment followed by a second retention or receiving profile.
Acoustic latch	A time-stamped transition marker in the trace.
JSF / JDS	The inherited two form-field channel names and their mixing rule.

The metronome as an event cycle

The source's *Self-Stabilizing Metronome* is represented by repeated localization, execution, release and continuation:

$$s_0 \xrightarrow{e_0} s_1 \xrightarrow{e_1} s_2 \xrightarrow{e_2} \dots \quad (19.2)$$

Each accepted event writes its transition and current support. A later event can request a child module through a dormant bud, so recurrence and structural extension share the same event grammar.

The common educational reading

The predecessor's observe–name–represent–combine–explain sequence becomes: name the two field inputs, choose the weights, locate a site, apply the written gates and explain the successor record. The technical layer carries the coordinate transforms and packed fields; the human-facing layer states what the inputs and outputs represent.

These correspondences preserve the earlier vocabulary while the operational equations retain their declared types, domains and record fields.

Source base [S0], pp. 3, 5, 7–13, 15–16, 19–21; [S2], pp. 14–18. The correspondence table carries the inherited domain vocabulary into the new structure.

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The integrated formal manifest

One architecture record

Collect the system as

$$\mathfrak{M}_{\text{II}} := (X, D, E, \mathcal{H}, \mathcal{H}, \mathcal{I}, V_{25}, \Lambda, \mathcal{P}, \mathfrak{L}, \mathcal{J}, \mathcal{C}). \quad (20.1)$$

Here $\mathcal{H}$ is the symbolic hypothesis profile, $\mathcal{I}$ the typed interpreter, Λ the address map, $\mathcal{P}$ the paired pinion family, $\mathfrak{L}$ the growth grammar, $\mathcal{J}$ the demand scheduler and $\mathcal{C}$ the inherited capstone/anchor record.

The integrated event equation

The complete operation is organized by

$$\begin{aligned} \mathcal{Y}_n &= \text{Execute}_{\mathcal{I}, d_n}(x_n, e_n; \Lambda, \mathcal{P}, \mathfrak{L}, \mathcal{J}), \\ x^* &= \text{Assemble}(x_n, \mathcal{Y}_n), \\ d^* &= \text{RewriteFrom}_{d_n}(x_n, e_n, \mathcal{Y}_n), \\ (x_{n+1}, d_{n+1}) &= \text{Commit}_{s_n, e_n}(x^*, d^*; \mathcal{Y}_n). \end{aligned} \quad (20.2)$$

The commit uses the explicit branch in section 16. The source's paired kernel is preserved at the outside; the new field, topological transport, numerical stages and growth derivations occupy the work record inside.

The resulting operational roles

Spatial query. Locate an event in a declared chart, quantize its log-polar position and resolve its local payload through the binary comparator.

Paired field execution. Evaluate the two field channels, transport their orientation and apply the two declared elementwise gates.

Temporal continuation. Advance the chosen dynamic channels and write their W coordinate through the recorded integrator stages.

Demand-driven growth. Realize the next required five-by-five module from its seed and guarded productions before field access.

Symbolic governance. Record hypothesis evidence and commit the description that governs the following event.

Maktub II. The written state, written rule and written growth instruction meet in one event. The trefoil supplies their execution structure; the 25-site module supplies local form; the doubled pinion supplies paired transport and gating; the timeline and L-system supply continuation.

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Notation and naming register

Symbol or term	Meaning in this edition
x_n, d_n, e_n	Committed state, current operator description, incoming event.
$\mathcal{K}, \mathcal{K}_{\Pi}, \mathcal{P}_n$	Paired kernel, extended transaction and packed pinion family.
$z = (p, W, \xi)$	Spatial coordinates, one-dimensional time, additional dynamic channels.
k_1, k_2, k_3, k_4	RK4 stage vectors; h_n is the event's step size.
$\mathcal{Y}_n, \mathcal{T}_n$	Event work record and committed trace.
$\mathcal{H}_n, \mathcal{R}_n$	Hypothesis family and outcome/evidence records.
$v, \chi, (i, j), k$	Module path, chart identifier, five-by-five indices, integer site key.
r, ρ, θ, β	Radius, log-radius code, azimuth and elevation.
o, E, A, Δ_n	Origin, committed eigenframe, frame matrix and its eigenvalue diagonal.
Λ	The global coordinate-to-address map in the manifest.
$\mathbb{K}^A, \mathbb{K}^B$	Paired Klein topological carriers.
κ, ε, J	Channel coordinate, orientation bit and chart tangent transport.
$\Omega_\sigma, \delta_\sigma$	A declared bounded-region profile and its signed-distance field.
F_σ, g_σ	A named stored field and its gate function.
$\odot, G^{(1)}, G^{(2)}$	Elementwise multiplication and the two successive gate arrays.
$H, M, \mathcal{C}, a_{\text{out}}$	Normalized Hadamard transform, sign matrix, capstone and external anchor.
$N, \alpha(N)$	Positive model quantity and its mixture weight.
φ_g, ψ_g	$(1 + \sqrt{5})/2$ and its positive reciprocal.
$\mathfrak{L}, \omega, \mathcal{P}_L$	Parametric grammar, word and production set.
T, H, B, P, A in grammar	Seed, header, group, payload and dormant bud; typed grammar symbols.
Need, Ready, $\mathcal{J}$	Demand predicate, availability predicate and scheduling rule.
L_B, Γ, Q	Bond Laplacian, release rates and growth-transfer matrix.
$\rightsquigarrow$	Domain correspondence or naming correspondence.

The grammar letters are symbolic tokens, the Hadamard H is a matrix, and the hypothesis family is $\mathcal{H}$. The surrounding type identifies each use. Units, parameter values and serialization names are fixed in the runtime profile.

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Source register and edition lineage

[S0] **Predecessor formalization.** Maktub_Tom_Klootwijk_Formalized_Paradigm(1).pdf. 22 pages. *Maktub: Formal specification of the paradigm*, TK-MAKTUB-2026-01, version 1.0, 09 October 2026. Supplies the paired kernel, mirror, capstone, log-polar field, two-field mixture and source-domain register.

[S1] **Repository and trefoil progression.** so I Tom Klootwijk NL200678942 10.pdf. 5 pages. Supplied conversation record. Its corrected third repository name is TOM-GR0K; the three-role composition appears on p. 4.

[S2] **Five-star and packed-pinion development.** 5 star if stars topology which is a log encoded polar LUT binary search tree because 5.pdf. 18 pages. The progression is: five-by-five topology and lookup (pp. 1–6); spatial-time extension and RK4 (pp. 6–8); doubled Klein/SDF/Hadamard packing and phi/psi (pp. 8–13); bonds, distribution and L-system growth (pp. 14–18).

[S3] **Progression with the just-in-time addition.** so I Tom Klootwijk NL200678942 109.pdf. 6 pages. Extends the supplied repository/trefoil conversation with the topology addition and its just-in-time reading on pp. 5–6.

Mathematical reference conventions

[M1] Przemyslaw Prusinkiewicz and Aristid Lindenmayer, *The Algorithmic Beauty of Plants*, chapter 1, sections 1.2 and 1.10. Algorithmic Botany author-hosted edition. Parallel and parametric rewriting conventions.

[M2] Y. Q. Wang, *Runge-Kutta method*, Math 4513 teaching notes, Oklahoma State University. Classical four-stage update; this edition writes stage slopes with the step size factored into the final combination.

[M3] Jeremy Orloff, *Symmetric Matrices*, ES.1803, MIT OpenCourseWare, Spring 2024. Real symmetric eigenframe convention.

[M4] Cornell Mathematics, *The Klein Bottle and a Mystery Surface*, Math Explorers' Club materials. Square-side identification and reflected continuation.

Edition contribution and attribution

This edition defines the typed extended state, the 25-key quantizer and search recurrence, the paired-channel normalizer, the two-gate composition, the golden-scale placement profile, the guarded growth productions, the demand scheduler, the transfer ledger and the complete event transaction. These definitions compose the structures named across [S0]–[S3].

The earlier source PDFs named inside [S0] remain part of its inherited provenance. The four supplied documents above are the direct source corpus for this new edition.

Paradigm author: Tom Klootwijk

NL200678942 / 10-07-1990

Maktub II: Trefoiled state, symbolic learning and just-in-time growth

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